PRST Class (22/10/2020)

Geometric distribution :

Consider series of Bernoulli trials. (success prob p)

X - number of failures before the 1st success.

$\left(\begin{matrix} 1 & 2 & 3 & 4 \\ F & F & F & S \\ S & & & \end{matrix}\right.$ $\quad X=3$

$X=0$

$\begin{matrix} 1 & 2 & 3 \\ F & S & \end{matrix}$ $\quad X=1$

X can take values $0,1,2,3,\ldots$

$P(X=x) = pq^x \qquad x = 0,1,2,3,\ldots$

$P(X=0) = p \qquad q = 1-p$

$P(X=1) = pq$

$P(X=2) = q^2 p$

$\left(\underbrace{G\ G\ G \ldots}_{X} D \quad E(X) = 200 \right)$

$P(X=x) = pq^x \qquad E(X) = \sum x pq^x$

$\sum_{x=0}^{\infty} (pq^x) = 1 \qquad 0 < p,\ q < 1$

$\sum_{x=0}^{\infty} p(1-p)^x = 1$

$\Rightarrow \frac{d}{dp} \sum_{x=0}^{\infty} p(1-p)^x = 0$

$\Rightarrow \sum^{\infty} \frac{d}{} \left(p(1-p)^x\right) = 0$

$$\Rightarrow \sum_{x=0}^{\infty} \frac{d}{dp}\left(p(1-p)^{x}\right) = 0$$

$$\Rightarrow \sum_{x=0}^{\infty}\left[(1-p)^{x} - p\,x\,(1-p)^{x-1}\right] = 0$$

$$\Rightarrow \underbrace{\sum_{x=0}^{\infty}(1-p)^{x} - \sum_{x=0}^{\infty}\frac{x p q^{x}}{q}}_{} = 0$$

$$\Rightarrow \frac{1}{p} - \frac{1}{q}E(X) = 0$$

$$\Rightarrow E(X) = \frac{q}{p}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = ?$$

$$\sum_{x} p q^{x} = 1$$

$$\sum_{x}(1-p)^{x} - \sum_{x} xp(1-p)^{x-1} = 0$$

$$\Rightarrow -\sum_{x} x(1-p)^{x-1} - \sum\left[x(1-p)^{x-1} - x(x-1)p(1-p)^{x-2}\right] = 0$$

$$\Rightarrow -\sum_{x} x q^{x-1} - \sum x q^{x-1} + \sum x(x-1)pq^{x-2} = 0$$

$$\Rightarrow \sum x(x-1)pq^{x-2} = 2\sum x q^{x-1}$$

$$\Rightarrow \frac{1}{q^2}\sum x(x-1)pq^{x} = \frac{2}{pq}\sum_{x} xpq^{x}$$

$$\Rightarrow \frac{1}{q^2}E(X(X-1)) = \frac{2}{pq}E(X) = \frac{2}{p\cancel{q}}\frac{\cancel{q}}{p} = \frac{2}{p^2}$$

$$E(X(X-1)) = \frac{2q^2}{p^2}$$

$$\Rightarrow \frac{1}{q^2} \quad E(X(X-1)) = \frac{2q^2}{p^2}$$

$$\Rightarrow$$

$$E(X^2) = E(X(X-1)) + E(X)$$

$$Var(X) = \frac{q}{p^2}$$

Eg. A company is producing ~~go~~ electric bulb and 90%. products are good products. Conduct a ~~test~~ quality testing. What is the expected number of good bulb before getting ~~a def~~ the 1st defective bulb before getting the 1st defective bulb?

Sol: X: # goods bulbs before getting the 1st defective bulb.

$$P(X = x) = (.9)^x (.1)$$

$$E(X) = \frac{.9}{.1} = 9$$

Negative Binomial distribution:

r = 3

FF S FFFF S F S $\quad X = 7$

FF FF SSS $\quad X = 4$

SSS $\quad X = 0$

Consider indefinite series of Bernoulli ... prob).

Consider indefinite [illegible] trials. (p – success prob).

X: number of failures to get ~~r~~ 1st r successes

$$P(X=x) = \binom{x+r-1}{x} p^r q^x$$

$x = 0, 1, 2, \ldots$

$r = 2$ $\quad x = 5$

FFSFFFS

FFFFFSS

SFFFFFS

$\vdots$

$$\binom{x+r-1}{x} = \binom{x+r-1}{r-1}$$

$$\binom{n}{r} = \binom{n}{n-r}$$

$$E(X) = \frac{rq}{p}$$

$$Var(X) = \frac{rq}{p^2}$$

$$(a+b)^{-n}$$